



CST8508 – Machine Vision – Project

Learning outcomes

The project's goal is to track a person visible from your webcam and ensure that they are centered in the image.

Instructions

- Establish a way of streaming images from your webcam as we did in lab 5. If you do not have access to a GPU on your local machine, and you need to use Google Colab, you can refer to the provided notebook that shows how to stream from a camera on Colab.
- For every frame:
 - You will run an object detection model to get the bounding boxes and confidences. You will use a pre-trained model from mmdetection for that. You will be using a pre-trained **rtmdet-s** model. Pre-trained model and corresponding config can be found [here](#).
 - We only want predictions that belong to class person. As such, you can filter out all other detections. The person class has label “0” in the output of rtmdet.
 - Once you have the bounding boxes of people and corresponding confidences, run a tracker. You might find it easier to plug in a tracker without using mmtracking. For example, you can use [this](#) simpler implementation of bytetrack. You just have to install that repo and then you can call bytetrack and use it on the outputs of your detector.
 - At this point, your solution must be able to stream from the camera, and display the tracked objects on screen (display the detection and its corresponding ID)
 - You can build your system under the assumption that there is only ONE person detected in an image. If you detect more than one person, just consider the person with the largest bounding box area.
 - Now, for that tracked person, you need to check if they are centered in the image or not. To do that, you will:
 - Get the center of the image (x coordinate is width/2, y coordinate is height/2)

- Get the center of the tracked person bounding box (x coordinate is box top left x + (box width)/2, y coordinate is box top left y + (box height)/2).
- If box center x coordinate is to the left of the center, print a message in green saying that the camera must move to the left. Otherwise, if it is to the right of the center, print a message in green saying that the camera must move to the right.
- If box center y coordinate is higher than the center, print a message in red saying that the camera must move up. Otherwise, if it is below the center, print a message in reds aying that the camera must move down.
- For each printed message, you must also state by how much the camera must move in a certain direction.

What to submit on Brightspace

- Sample demo video of you testing the system with your webcam
- All python code written
- Config files used
- Report showing your evaluation analysis and lessons learned.

Or

Project Overview

For your final project, you will select a dataset from Kaggle and choose a project of your choice. This document outlines the requirements and deliverables for your project submission.

Project Requirements

1. Dataset Selection:

- Choose a dataset from Kaggle that interests you and is suitable for your project.
- Ensure the dataset is comprehensive enough to allow for meaningful analysis and insights.

2. Project Scope:

- Define the scope of your project clearly. This includes the problem statement, objectives, and expected outcomes.
- Ensure the project is feasible within the given timeframe and resources.

Deliverables

You are required to submit the following items on Brightspace:

1. Sample Demo Video:

- Create a short demo video (3-5 minutes) showcasing your project.
- The video should include a demonstration of your project in action, highlighting key features and functionalities.
- Ensure the video is clear and easy to follow.

2. Python Code:

- Submit all the Python code you have written for the project.
- Ensure your code is well-documented with comments explaining the purpose of each section.
- Organize your code into logical modules and functions for readability and maintainability.

3. Configuration Files:

- Include any configuration files used in your project (e.g., .env files, JSON/YAML configuration files).
- Ensure these files are properly formatted and contain all necessary settings for running your project.

4. Report:

- Write a comprehensive report detailing your project.
- The report should include the following sections:
 - **Introduction:** Briefly describe the project, including the problem statement and objectives.

- **Dataset:** Provide an overview of the dataset you selected, including its source, structure, and any preprocessing steps you performed.
- **Methodology:** Explain the methods and techniques you used to analyze the data and achieve your project objectives.
- **Results:** Present the results of your analysis, including any visualizations, metrics, or insights you obtained.
- **Evaluation:** Evaluate the performance of your project, discussing any challenges you faced and how you addressed them.
- **Lessons Learned:** Reflect on what you learned during the project, including any improvements you would make in the future.

Submission Guidelines

- Ensure all deliverables are submitted on Brightspace by the specified deadline.
- Double-check that all files are correctly named and organized.
- Review your work for completeness and accuracy before submission.

Additional Tips

- Start early to give yourself ample time to explore the dataset and refine your project.
- Utilize online resources and documentation to enhance your project.

Good luck with your final project! If you have any questions or need further assistance, feel free to reach out.